1 0.00074 7.39559

Residual 86.10 10.90 2.70 0.30 71.7500 109 0.8438 0.01176 117.5958

Distillate 85.05 13.42 1.50 0.03 70.8750 134.2 0.4688 0.006614 66.1376

Residual 87.17 11.03 1.50 0.30 72.6417 110.3 0.4688 0.006453 64.5291

* Based on properties in the IMO NOx Monitoring Guidelines, resolution MEPC.103(49).

Table 2: Emissions calculations corresponding to 1.5 % fuel sulphur

CO2 SO2 Exh SO2/CO2 Exh S/C

% ppm4 ppm4/% m/m

Distillate 0.17% S 8 59.1 7.4 0.00197

Residual 2.70% S 8 939.7 117.5 0.03136

Distillate 1.5% S 8 528.5 66.1 0.01764

Residual 1.5% S 8 515.7 64.5 0.01721

Distillate 1.5% S 0.5 33.0 66.1 0.01764

Residual 1.5% S 0.5 32.2 64.5 0.01721

-

Residual 2.70% S 8 939.7 117.5 0.03136

Distillate 1.5% S 8 528.5 66.1 0.01764

Residual 1.5% S 8 515.7 64.5 0.01721

Distillate 1.5% S 0.5 33.0 66.1 0.01764

Residual 1.5% S 0.5 32.2 64.5 0.01721

SO2/CO2 ratio vs % sulphur in fuel

0

50

100

150

200

250

0,00 0,50 1,00 1,50 2,00 2,50 3,00 3,50 4,00 4,50 5,00

% m/m sulphur in fuel

SO2(ppm)/CO2(%)

Distillate fuel

Residual fuel

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4 Correspondence between 65 (ppm 4/%) SO2/CO2 and 6.0 g/kWh is demonstrated by showing that their S/C ratios are similar. This requires the additional assumption of a brake-specified fuel consumption value of 200 g/kWh. This is an appropriate average for marine diesel engines. The calculation is as follows:

S/Cfuel = brake-specific SO2

(MWS / MWSO2)

BSFC

(% carbon in fuel / 100)
brake-specific SO2 = 6.0 g/kW-hr
MWS = 32.065 g/mol
MWSO2 = 64.064 g/mol
BSFC = 200 g/kW- hr
% carbon in 1.5% sulphur fuel (from table 1) = 85.05% (distillate) or 87.17%
-
(MWS / MWSO2)
BSFC
(% carbon in fuel / 100)
brake-specific SO2 = 6.0 g/kW-hr
MWS = 32.065 g/mol
MWSO2 = 64.064 g/mol
BSFC = 200 g/kW- hr
% carbon in 1.5% sulphur fuel (from table 1) = 85.05% (distillate) or 87.17%
(residual)
S/Cresidual fuel = 6.0
(32.065 / 64.064) = 0.01723 200
(87.17% / 100)
S/Cdistillate fuel = 6.0
(32.065 / 64.064) = 0.01765 200
(85.05% / 100)
Note 1: The S/C mass ratios calculated above, based on 6.0 g/kWh and 200 g/kWh BSFC,
are both within 0.10% of the S/C mass ratios in the emissions table (Table 2).
Therefore, 65 (ppm4/%) SO2/CO2 corresponds well to 6.0 g/kWh.
Note 2: The value of 6.0 g/kWh, hence the 200g/kWh brake-specified fuel consumption is
taken from MARPOL Annex VI as adopted by the 1997 MARPOL Conference.
5 Thus, the working formulas are as follows:
For complete combustion = SO2 (ppm*) ■ 65 CO2 (%*)
For complete combustion = SO2 (ppm*) ■ 65 CO2 (%*) + (CO(ppm*)/10000) + (THC(ppm*)/10000)
* Note: gas concentrations must be sampled or converted to the same residual water
content (e.g., fully wet, fully dry).
-
For complete combustion = SO2 (ppm*) ■ 65 CO2 (%*) + (CO(ppm*)/10000) + (THC(ppm*)/10000)
* Note: gas concentrations must be sampled or converted to the same residual water
content (e.g., fully wet, fully dry).
6 The following is the basis of using the (ppm 4/%) SO2/CO2 as the limit for determining
compliance with regulation 14.1 or 14.4 of MARPOL Annex VI:
.1 This limit can be used to determine compliance from fuel oil burners that do
not produce mechanical power.
.2 This limit can be used to determine compliance at any power output,
including idle.
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.3 This limit only requires two gas concentration measurements at one sampling location.
.4 There is no need to measure any engine parameters such as engine speed, engine torque, engine exhaust flow, or engine fuel flow.
.5 If both gas concentration measurements are made at the same residual water content in the sample (e.g., fully wet, fully dry), no dry- to-wet
-
speed, engine torque, engine exhaust flow, or engine fuel flow.
.5 If both gas concentration measurements are made at the same residual water content in the sample (e.g., fully wet, fully dry), no dry- to-wet
conversion factors are required in the calculation.
.6 This limit completely decouples the thermal efficiency of the fuel oil combustion unit from the EGC unit.
.7 No fuel properties need to be known.
.8 Because only two measurements are made at a sing